

Predictive Modeling of Heat Exchanger Fouling Factor

CH-512: Machine Learning in Chemical Engineering

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Goal : Develop an optimal soft sensor for the Fouling Factor using multivariate regression and advanced nonlinear techniques.

Fouling in Heat Exchangers

Heat Exchangers (HEs) are fundamental unit operations, but their performance is severely degraded by Fouling the accumulation of solid deposits on heat transfer surfaces.

What is Fouling?

Deposition of undesired materials on the heat transfer surface, leading to reduced heat transfer efficiency and increased maintenance.

The Target Variable:

Our objective is to model and predict the Fouling Factor (R_f) in real-time. R_f is a complex, non-linear function of multiple operating conditions (Time, T_s , v , etc.).

Direct Measurement of R_f is Impractical. We develop a Soft Sensor (Virtual Sensor) using Machine Learning to provide a continuous, accurate estimate of R_f from seven easily-measured input variables. This is crucial for Predictive Maintenance and Process Optimization.

The process is governed by competing mechanisms (deposit initiation, transport, attachment, and removal), mandating the use of advanced techniques like Principal Component Regression (PCR) and Artificial Neural Networks (ANN) to capture the complexity.

Linear Regression (Correlation Coefficient)

R^2 (degree-1): 0.4648

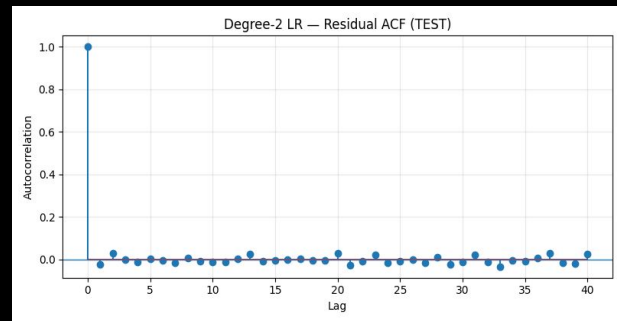
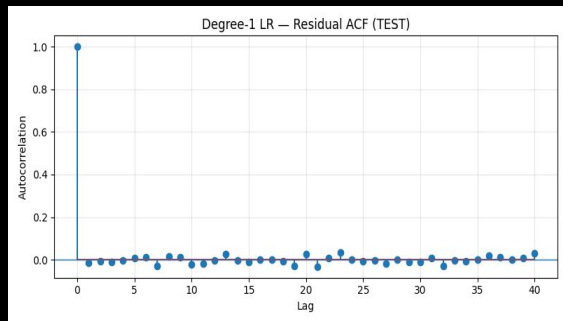
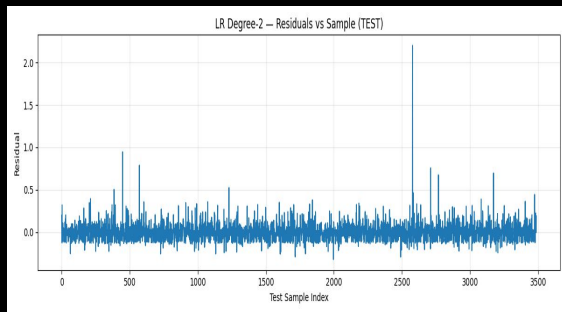
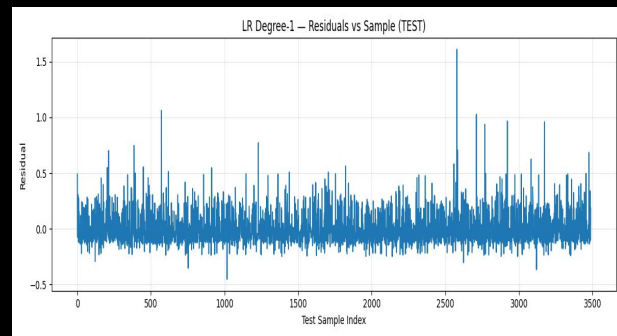
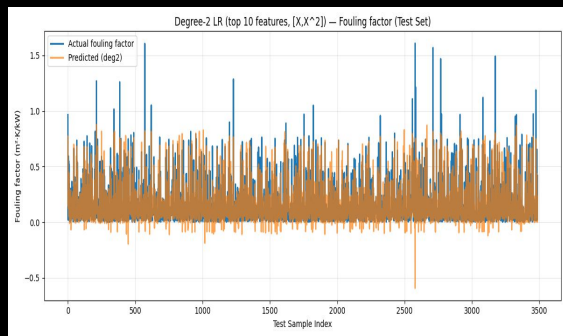
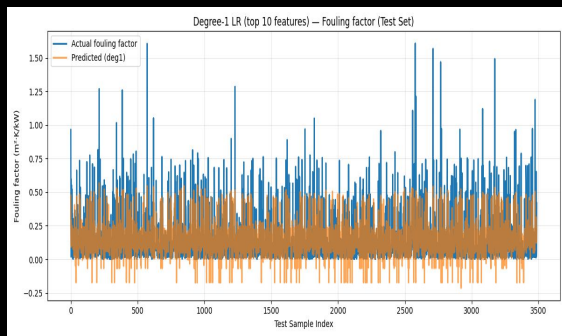
R^2 (degree-2): 0.6983

SSE: 36.884725, AIC: -5947.40, BIC: -5879.67

Residual ACF (first few):

Degree-1: [1. -0.01594759 -0.00716343 -0.00955203 -0.0022501]

Degree-2: [1.0e+00 -2.13920788e-02 2.98410e-02 -2.2892e-05 -1.064e-02]



Lasso Regression

R^2 (degree-1): 0.4136

R^2 (degree-2): 0.6988

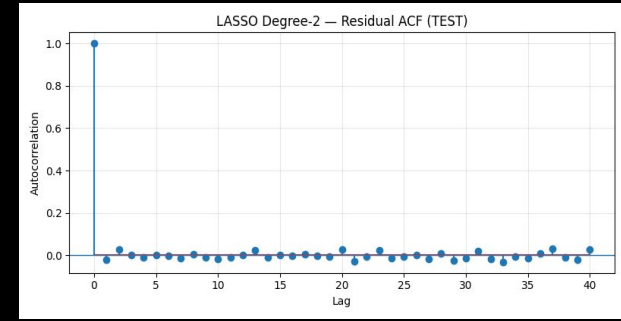
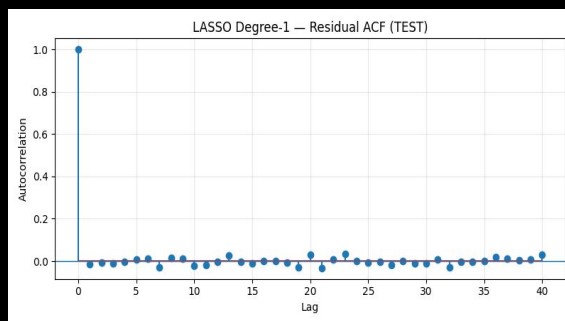
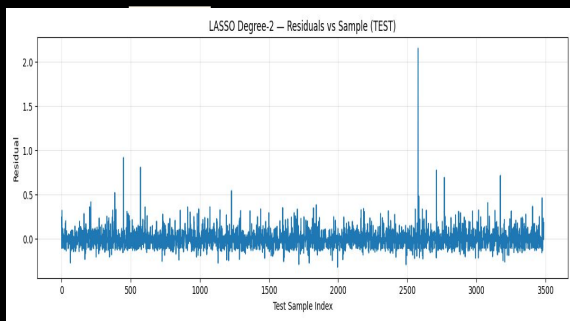
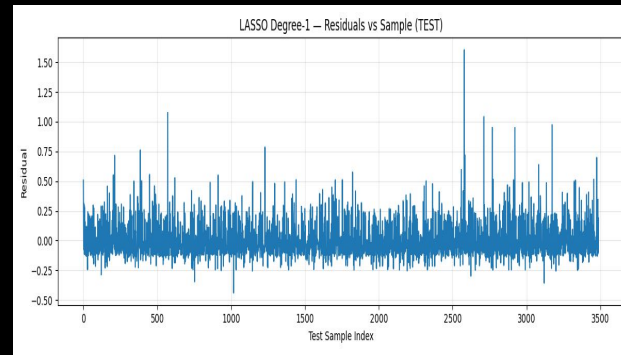
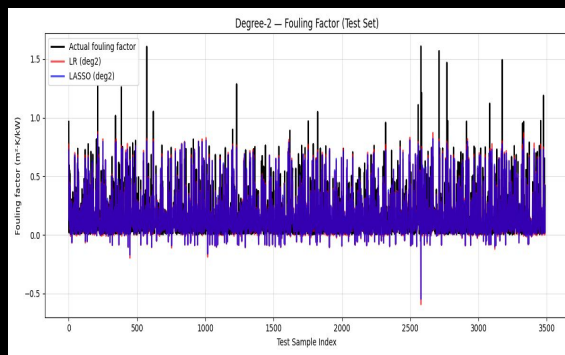
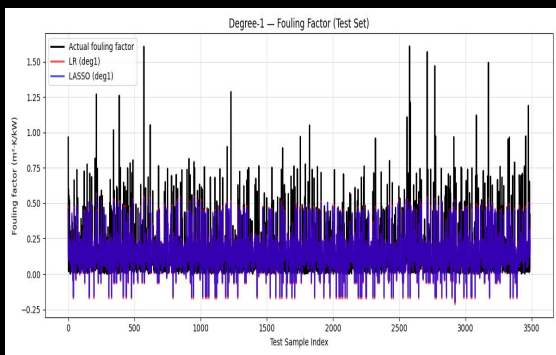
SSE: 68.841130, AIC: -3780.87, BIC: -3743.93

SSE: 69.148944, AIC: -3765.31, BIC: -3728.37

Residual ACF (first few):

LASSO Deg1: [1. -0.01495717 -0.00766624 -0.01050332 -0.00256428]

LASSO Deg2: [1.00e+00 -2.16082579e-02 2.74326050e-02 5.38265477e-04 -1.02652632e-02]



Linear Regression (PCA)

R^2 (degree-1): 0.4369

R^2 (degree-2): 0.7436

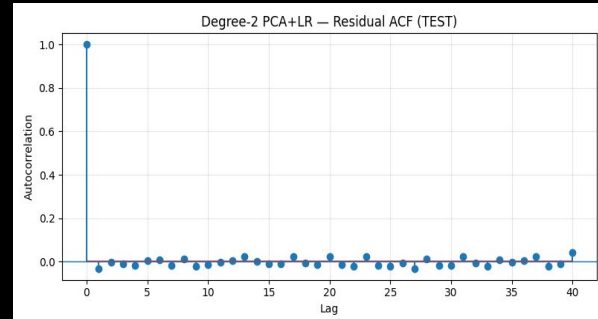
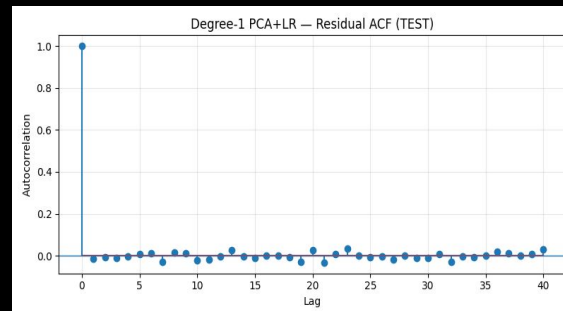
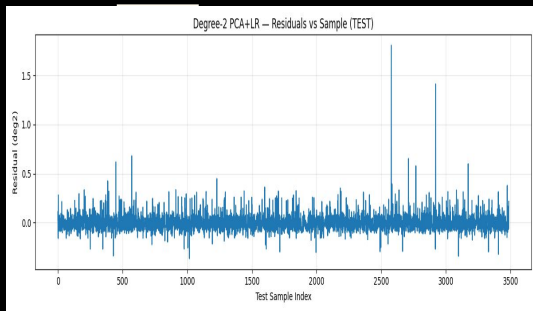
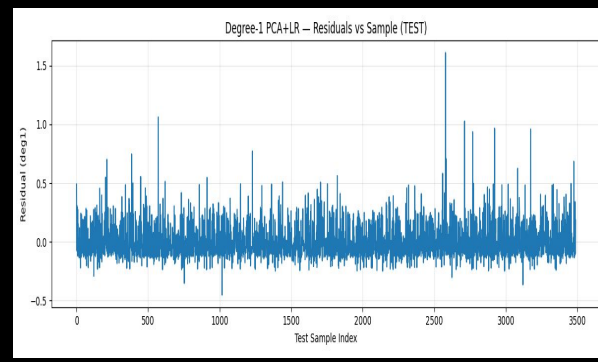
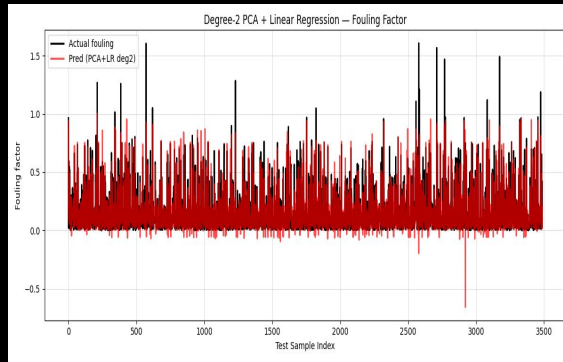
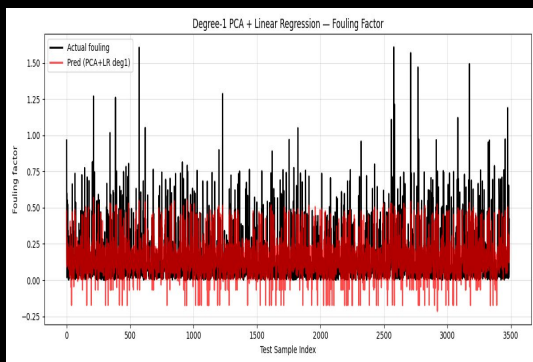
SSE: 65.432611, AIC: -3957.99, BIC: -3921.05

SSE: 36.884725, AIC: -5947.40, BIC: -5879.67

Residual ACF (first few):

Degree-1: [1. -0.01146593 -0.00271934 -0.01285285 -0.0079183]

Degree-2: [1.00e+00 -2.13920788e-02 2.98402610e-02 -2.28908052e-05 -1.06420390e-02]



Ridge Regression

R^2 (degree-1): 0.4648

R^2 (degree-2): 0.6983

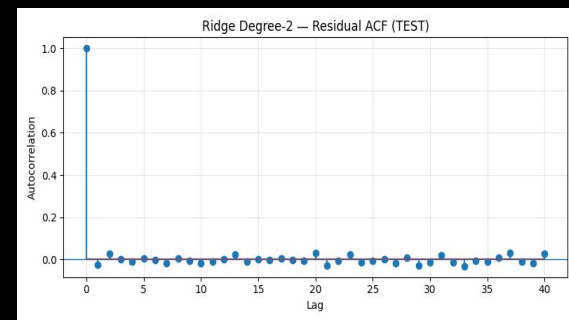
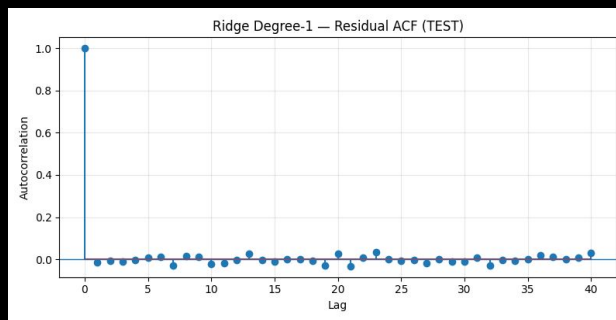
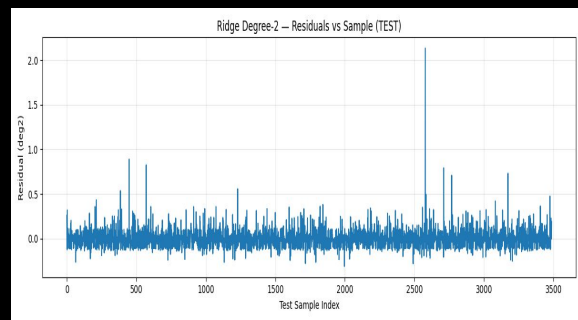
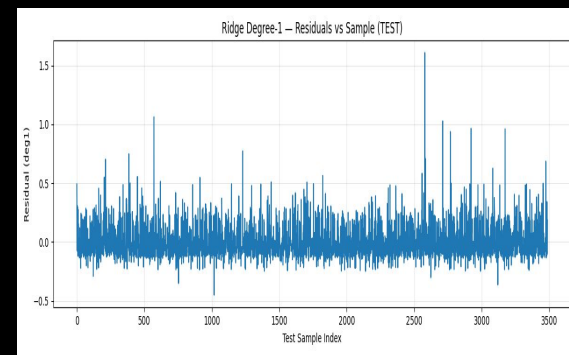
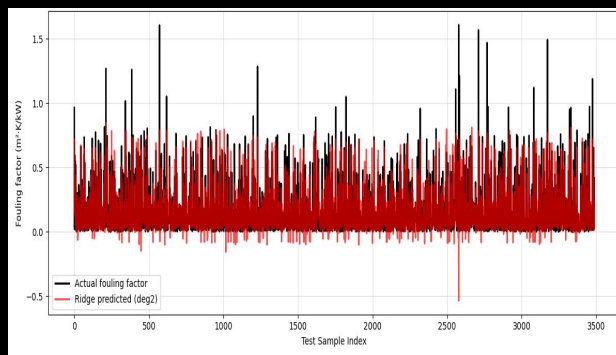
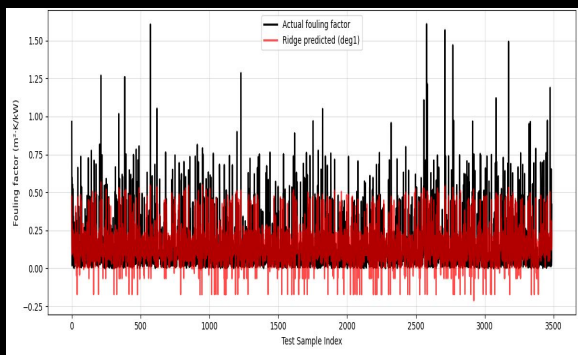
SSE: 68.858405, AIC: -3779.99, BIC: -3743.05

SSE: 36.816255, AIC: -5953.88, BIC: -5886.15

Residual ACF (first few):

Ridge deg1: [1. -0.01589594 -0.00720072 -0.00961515 -0.00228428]

Ridge deg2: [1. -2.43756054e-02 2.72746695e-02 2.97586196e-05 -8.32338630e-03]



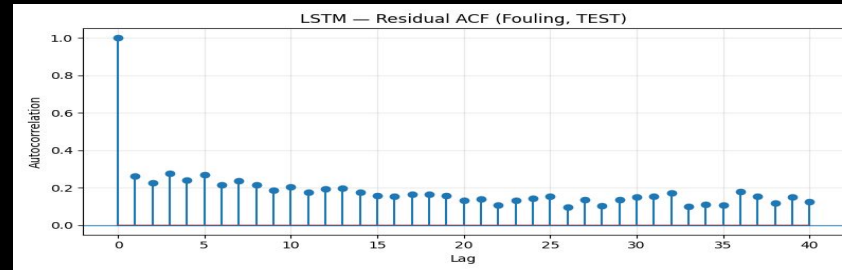
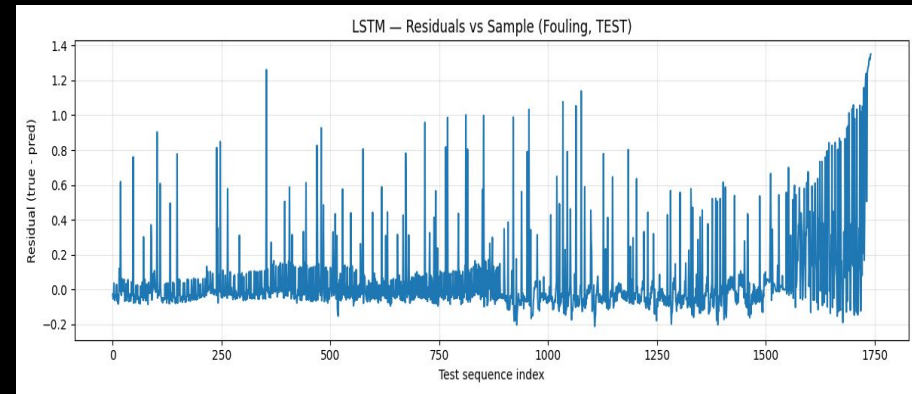
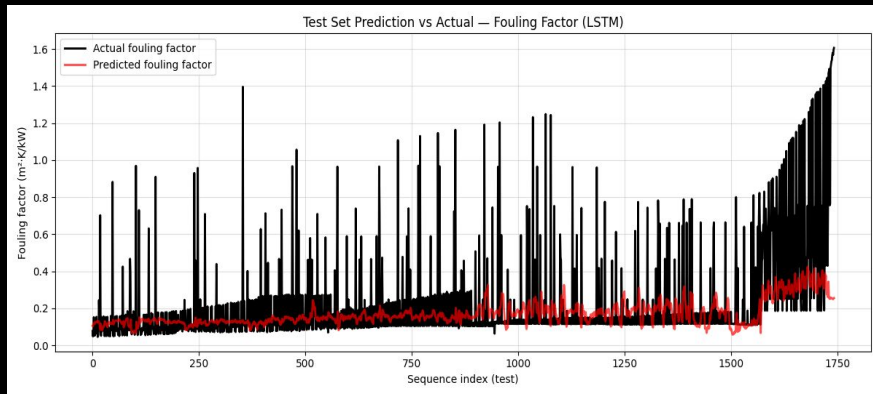
LSTM

$R^2 : 0.1124$

SSE=96.970200, AIC=36394.01, BIC=136040.75

Residual ACF (first few):

[1. 0.26333523 0.22732589 0.27542707 0.24076022]



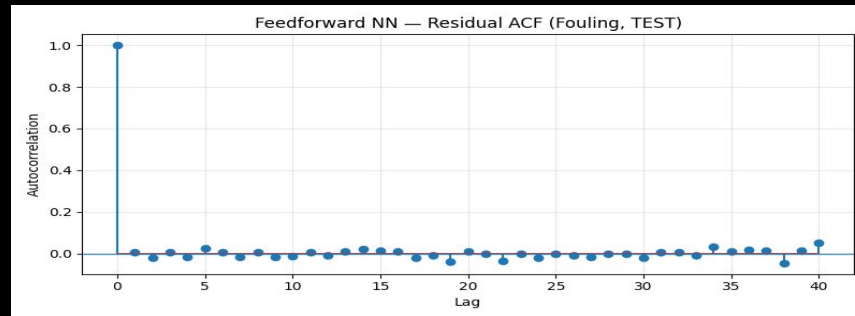
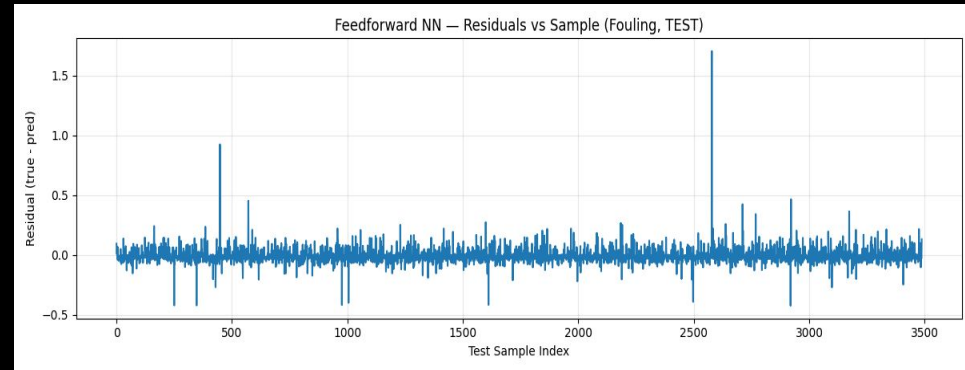
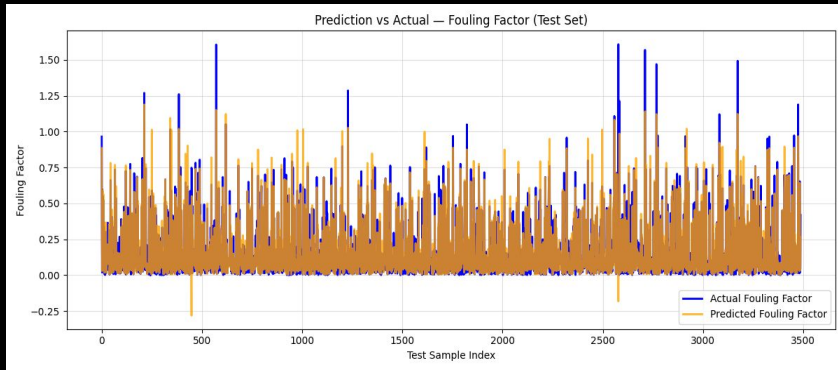
Conventional Neural Network

$R^2 : 0.8723$

$SSE=15.616253$, $AIC=-8197.28$, $BIC=-5826.80$

Residual ACF (first few lags):

[1.00e+00 -1.32860721e-03 -1.99832357e-02 3.65210726e-04 -2.12069079e-02]



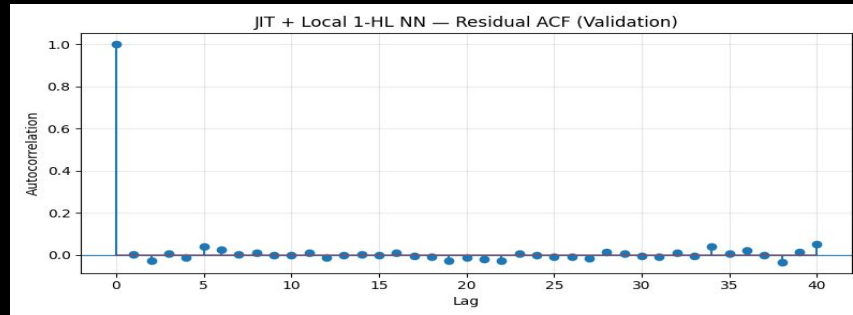
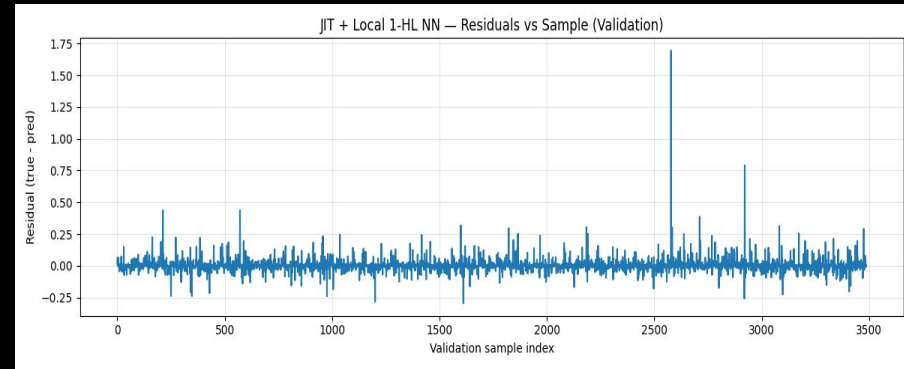
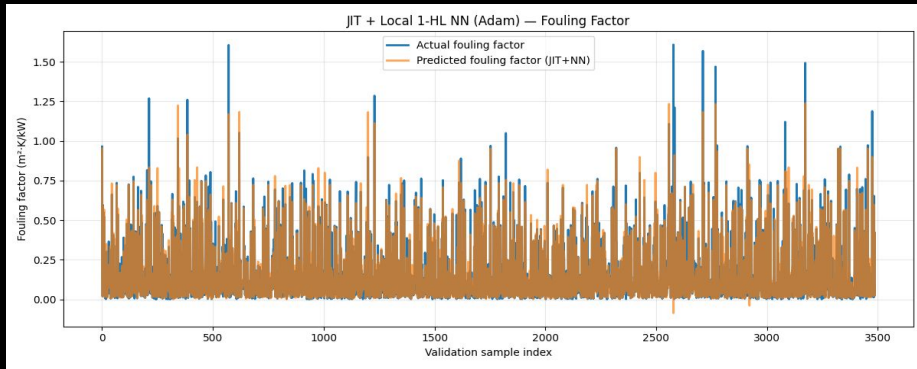
JIT (Neural Network Based)

R^2 : 0.9143

SSE: 11.241284, AIC: -9999.85, BIC: -9648.90

Residual ACF (first few):

[1. 0.00177211 -0.02699628 0.00785532 -0.01081015]



Autoregressive based Deep NN Model

R^2 (degree-1): 0.4402

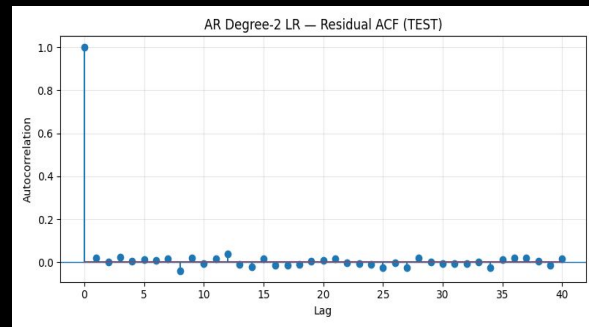
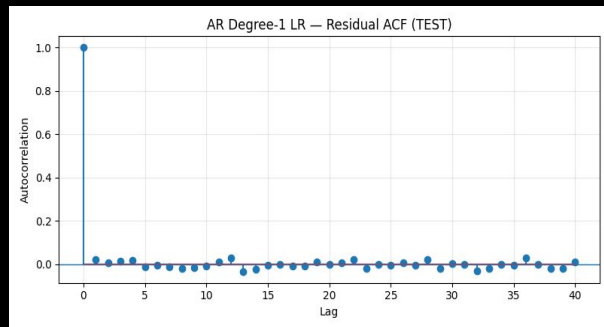
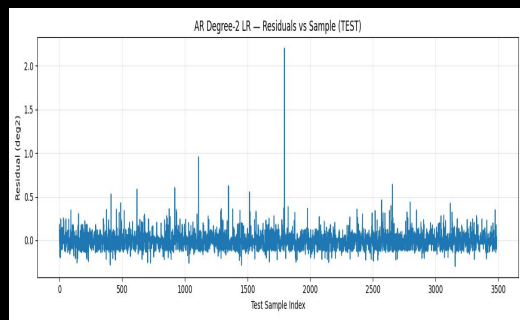
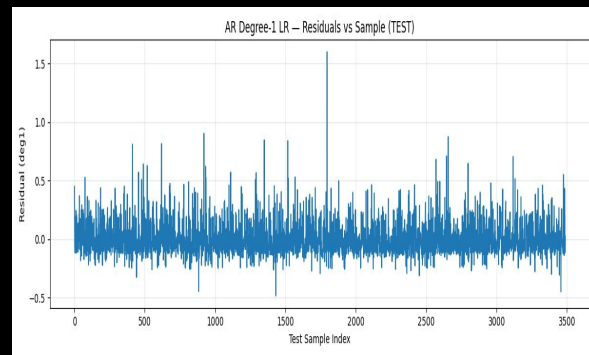
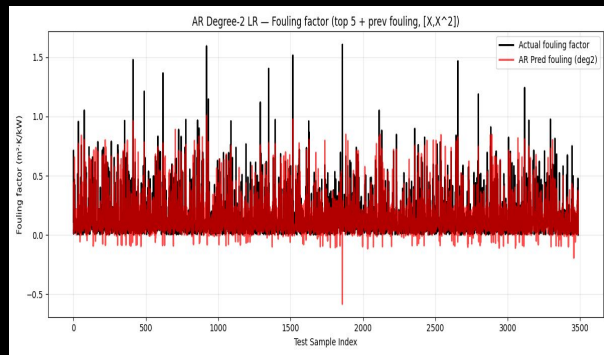
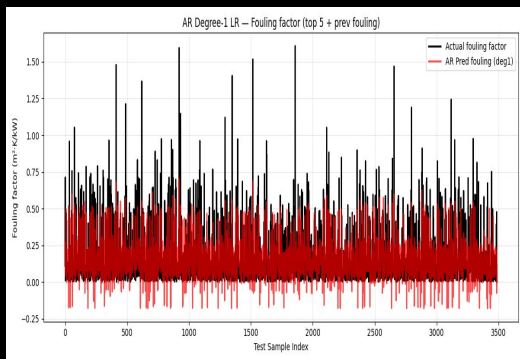
R^2 (degree-2): 0.6893

SSE: 68.586666, AIC: -3791.79, BIC: -3748.69

Residual ACF (first few):

Degree-1: [1 0.0223805 0.00608725 0.01368736 0.01909037]

Degree-2: [1 0.01921243 0.0019861 0.02331847 0.00455201]



Conclusion

The core project goal, to develop an accurate, data-driven virtual sensor for the Fouling Factor (R_f) was successfully achieved using multivariate regression and neural network techniques.

- The Optimal Predictor is the Neural Network Based JIT model • This model achieved the highest predictive power on unseen data: Test R^2 : 0.9173.
 - It minimized the overall error while being reasonably penalized for complexity: Lowest Test SSE: 11.24
 - The substantial improvement in predictive performance shown by both the Second-Order Polynomial Regression and the ANN over the basic Linear Regression model confirms the highly non-linear dependence of the Fouling Factor on process variables (Temperature, Time, Velocity).
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- The validated soft sensor provides continuous, real-time R_f estimation, which is superior to traditional, time-consuming methods. This allows operators to:
 - Schedule Cleaning: Based on predicted fouling limits, minimizing unnecessary downtime.
 - Optimize Operating Conditions: To actively mitigate the rate of fouling accumulation.

Key Learnings

These learnings highlight the key technical and theoretical insights gained during the development of the soft sensor model.

- **Non-Linearity is Essential for Fouling Kinetics:**The significant performance gap (higher R^2 , lower SSE) between the Linear PCR model and the Polynomial PCR / Artificial Neural Network (ANN) models confirms that the Fouling Factor (R_f) is driven by complex, non-linear kinetics (e.g., deposition rate, competing removal mechanisms). Linear assumptions are inadequate for accurate prediction.
- **PCA for Model Stability, Not Just Reduction:**Principal Component Analysis (PCA) was critical for stabilizing the regression models by transforming the highly correlated original features into orthogonal (uncorrelated) Principal Components. This prevents issues like multicollinearity and ensures reliable coefficient estimation in Principal Component Regression (PCR).
- **The True Value of Information Criteria (AIC/BIC):**Metrics like R^2 prioritize fit, but AIC and BIC were vital for guiding model selection. They quantified the complexity penalty associated with the massive number of parameters (p) in the second-order polynomial model, pushing the final choice toward the most parsimonious model that still offered high predictive accuracy on the test set.
- **Model Validation via Residuals:**Analyzing the Residuals and their auto-correlation was the ultimate step in model validation. A random residual distribution (centered at zero) and low auto-correlation confirmed that the chosen model successfully captured all the systematic information (both steady-state and time-dependent) in the data.

Improvement Suggestions

The course has been helpful, yet I feel that a few thoughtful improvements could make it even more effective for students.

- Include more hands-on coding sessions using Python, TensorFlow, and Scikit-learn to support practical learning.
- The exam should be less calculation-heavy and focus more on conceptual reasoning and interpretation.

THANK YOU

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